Systems of Linear Equations and Matrices

Definition & Notation

1) Matrix is a rectangular array of elements arranged in rows and columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Row

Column

m = numbers of rows

n = numbers of columns

a_{jk} = any element (or entry)

$m \times n$ = order of matrix (or dimension of matrix)

Definition & Notation(Cont')

2) Shape of matrix

Eg:

$$A = \begin{bmatrix} 6 & 5 & 3 & -4 & 3 \\ -1 & -6 & 6 & 7 & 8 \\ 8 & 1 & -2 & 0 & 16 \end{bmatrix}$$

- Order of $A = 3 \times 5$
- $\boxed{a_{11} = 6}$, $\boxed{a_{12} = 5}$, $\boxed{a_{35} = 16}$, $\boxed{a_{34} = 0}$, these are some of the elements of A.
- Shape = Rectangle.

Algebraic operations with matrices

- = equality
- $\pm$ addition and subtraction
- $\times$ multiplication
- $\div$ division (not defined)
- Power (e.g.A^2, A^4)
- Multiplication by scalar
- Transpose

i. *Equality*

- Two matrices *A* and *B* are equal if they have same order and their corresponding elements are equal.

  $A = B \Leftrightarrow a_{ij} = b_{ij}$ for all *i* and *j*.

Eg:

$$A = \begin{bmatrix} x+y & 2 \\ 3 & x-y \end{bmatrix} \quad B = \begin{bmatrix} 4 & u \\ v & 2 \end{bmatrix}$$

Determine *x*, *y*, *u* and *v* if $A = B$.

Solution: *u* =2, *v*=3, *x*=3, *y* =1.

$x + y = 4$ ———— (1)

$x - y = 2$ ———— (2)

(1)+(2):

$2x = 6$

$x = 3$

$y = 4 - 3 = 1$

ii. Addition and Subtraction

The matrices to be added or subtracted must be of the same order. The corresponding elements in the matrices are added or subtracted.

Eg.1:

$$\begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} + \begin{bmatrix} -1 & 2 \\ 9 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 12 & 6 \end{bmatrix}; \begin{bmatrix} -1 & -2 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} -2 & 3 \\ -1 & 8 \end{bmatrix} = \begin{bmatrix} 1 & -5 \\ 4 & -3 \end{bmatrix}$$

Eg.2:

$$\begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} \pm \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix}$$

**Cannot be added or subtract because the order of two matrices are not same.

iii. Matrix-matrix multiplication

- Two rectangular matrices $A\ (m_1 \times n_1)$ and $B\ (m_2 \times n_2)$ conformable if $n_1 = m_2$.

- Multiplication of *A* and *B* is define only if *A*, *B* are conformable, i.e. **the number of columns in *A* equal the number of row in *B***.

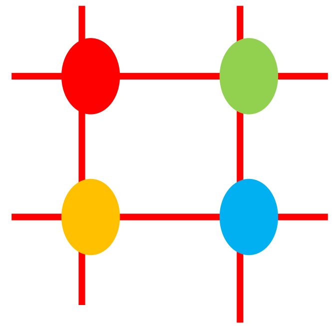
- The product of *AB* will be the matrix of order $(m_1 \times n_2)$ whose elements are equal to the sum of the product of the inner elements.

iii. *Matrix-matrix multiplication(Cont')*

- Eg.1:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{bmatrix}$$

(2 x 3) (3 x 2) (2 x 2)



- Eg.2:

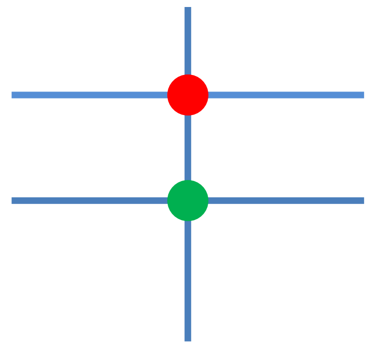
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 4 & 2 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} 1.1 + 2.4 + 3.5 & 1.0 + 2.2 + 3.1 \\ 2.1 + 0.4 + 1.5 & 2.0 + 0.2 + 1.1 \end{bmatrix} = \begin{bmatrix} 24 & 7 \\ 7 & 1 \end{bmatrix}$$

iii. *Matrix-matrix multiplication(Cont')*

Eg.2:

$$\begin{bmatrix}4 & 7 & 6\\ 2 & 3 & 1\end{bmatrix}\begin{bmatrix}8\\ 5\\ 9\end{bmatrix}=\begin{bmatrix}121\\ 40\end{bmatrix}$$

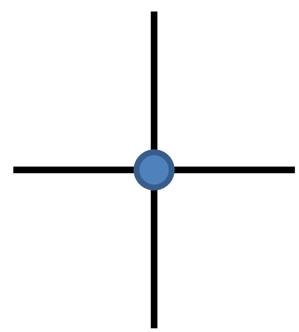
(2 X 3) (3 X 1) (2 X 1)



Eg.3:

$$\begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}3\\ 4\end{bmatrix}=\begin{bmatrix}3+8\end{bmatrix}=\begin{bmatrix}11\end{bmatrix}=11$$

(1 X 2) (2 X 1) (1 X 1)

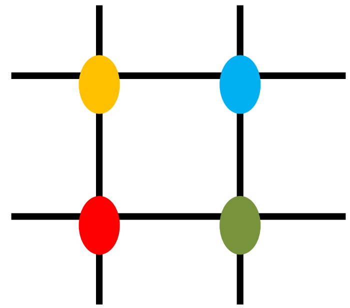


iii. Matrix-matrix multiplication(Cont')

Eg.4:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

(2 X 1) (1 X 2) (2 X 2)



Eg.5:

$$A = \begin{bmatrix} 2 & 4 & 5 \\ 3 & 9 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 3 & 5 \\ 8 & 7 & 1 \end{bmatrix}$$

$A^2 = AA$ is not defined.

AC is not defined also.

$$A^2 = \begin{bmatrix} 2 & 4 & 5 \\ 3 & 9 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & 5 \\ 3 & 9 & 1 \end{bmatrix}$$

(2 X 3) $\neq$ (2 X 3)

iii. Matrix-matrix multiplication(Cont')

- In general, $AB \neq BA$.
- A^2 is not necessary defined, except for A is a square matrix.
- $AB=0$ doesn't mean $A=0$ or $B=0$ or $BA=0$.

E.g:

$$A=\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix} \qquad B=\begin{bmatrix}-1 & 1\\ 1 & -1\end{bmatrix}$$

$$\left.\begin{array}{l} AB=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix} \\ \\ BA=\begin{bmatrix}1 & 1\\ -1 & -1\end{bmatrix}\end{array}\right\} \text{We see } A\neq 0, B\neq 0 \text{ or } BA\neq 0.$$

iv. Division

- We have A^{-1} equal to inverse of matrix buy not power negative one. The inverse of matrix will be discussed later.

$$A^{-1} \neq \frac{1}{A}$$

v. <u>*Positive power of matrix*</u>

- Only defined for square matrix.

$$A^n = AA^{n-1} = A^{n-1}A;\ A^m A^n = A^n A^m = A^{m+n}$$

E.g:

$A = \begin{bmatrix} 2 & -4 \\ -3 & 5 \end{bmatrix}$ we obtain $A^2 = \begin{bmatrix} 16 & -28 \\ -21 & 37 \end{bmatrix}$ $\quad A^3 = \begin{bmatrix} 116 & -204 \\ -153 & 269 \end{bmatrix}$

$$A^5 = A^3 A^2 = A^2 A^3 = \begin{bmatrix} 6140 & -10796 \\ -8097 & 14237 \end{bmatrix}$$

vi. *Multiplication by a scalar*

- The scalar *k* could be positive or negative.
- The product of scalar *k* and matrix $A(m \times n)$ yields an $(m \times n)$ matrix, each of whose elements are equal to the product of the scalar and the corresponding element of *A*.

$$A = \lfloor a_{ij} \rfloor, \ \ kA = \lfloor ka_{ij} \rfloor$$

E.g:

$$A = \begin{bmatrix} 8 & -2 & 14 \\ 6 & 16 & 12 \end{bmatrix}, \ 3A = \begin{bmatrix} 24 & -6 & 42 \\ 18 & 48 & 36 \end{bmatrix}, \frac{1}{2}A = \begin{bmatrix} 4 & -1 & 7 \\ 3 & 8 & 6 \end{bmatrix}$$

E.g: Multiply by negative unity will yield a negative matrix.

$$-A = \begin{bmatrix} -8 & 2 & -14 \\ -6 & -16 & -12 \end{bmatrix}$$ (negative of *A*)

vii. Transpose of a matrix

- The transpose A^T of a matrix *A* is formed by converting the rows of *A* to columns and its columns to rows, without disturbing their relative order.
- Thus the i^{th} row become the i^{th} column and the j^{th} column become the j^{th} row.

$$A = \lfloor a_{ij} \rfloor, \quad A^T = \lfloor a_{ji} \rfloor$$

E.g:

$$A = \begin{bmatrix} 3 & 9 \\ 2 & 6 \\ 4 & 1 \\ 8 & 5 \end{bmatrix} \qquad A^T = \begin{bmatrix} 3 & 2 & 4 & 8 \\ 9 & 6 & 1 & 5 \end{bmatrix}$$

Example

Let $A=\begin{bmatrix}2 & -1 & 1\\ 3 & 0 & -2\end{bmatrix}$, $B=\begin{bmatrix}1 & 3 & 0\\ 2 & -1 & 4\end{bmatrix}$. Find:

a) A^T and B^T.

b) $(3B)^T$

c) $(A+B)^T$

Some basic laws of Matrix Algebra

1. $AI = A = IA$ (where $I =$ identity matrix)
2. $A + 0 = 0 + A = A$ (0 is zero matrix)
3. $A + B = B + A$ Commutative Law
4. $A + (B + C) = (A + B) + C$ Associative Law

   $A(BC) = (AB)C$
5. $kA = Ak$ Commutative

   $(hk)A = h(kA)$ Associative Law

   $k(AB) = (kA)B = A(kB)$
6. $k(A + B) = kA + kB$ Distributive

   $(k + h)A = kA + hA$
7. $A(B + C) = AB + AC$ Distributive

   $(A + B)C = AC + BC$

Properties of transpose

1. $(A^T)^T = A$
2. $(AB)^T = B^T A^T$
3. $(A+B)^T = A^T + B^T$
4. $I^T = I$

Some special matrices

1. A matrix with one single row = row matrix (row vector). E.g: $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

2. A matrix with one single column = column matrix (column vector). E.g: $\begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$

3. Zero matrix all elements are zero.

E.g:

$$0 = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

Some special matrices (Cont')

4. Identity matrix :all diagonal elements are unity and zero for the rest.

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Some special matrices (Cont')

5. Diagonal matrix :only diagonal elements are non-zero.

$$\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

Some special matrices (Cont')

6. Symmetric matrix: $M^T = M$.

E.g:

$$M = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix} = M^T$$

$$B = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 2 & 0 & 0 & 3 \\ 1 & 0 & 1 & 0 \\ 3 & 3 & 0 & -1 \end{bmatrix} = B^T$$

Some special matrices (Cont')

7. An antisymmetric (or skew symmetric) matrix:

$$\boxed{M^T = -M.}$$

E.g:

$$C = \begin{bmatrix} 0 & 1 & -5 \\ -1 & 0 & 6 \\ 5 & -6 & 0 \end{bmatrix} \qquad C^T = \begin{bmatrix} 0 & -1 & 5 \\ 1 & 0 & -6 \\ -5 & 6 & 0 \end{bmatrix} = -C$$

E.g:

$$0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is both symmetric and anti-symmetric.

Note: 4. – 7. are square matrices.

Further Operation on Square Matrix

- (2 x 2) case.
- (3 x 3) and bigger dimension case.

minor, adjoint, cofactor, determinant, inverse.

(2 x 2) *case*

1. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

The determinant of *A* is $\boxed{\det A = |A| = ad - bc.}$

2. The inverse of *A* is

$$\boxed{A^{-1} = \frac{1}{|A|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}$$

(2 x 2) *case*(*Cont'*)

E.g. 1:

$$A = \begin{bmatrix} 4 & -3 \\ 1 & -6 \end{bmatrix}$$

$$|A| = ad - bc = 4(-6) - (-3)(1) = -24 + 3 = -21$$

$$A^{-1} = \frac{1}{|A|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{-21}\begin{bmatrix} -6 & 3 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} \frac{6}{21} & -\frac{3}{21} \\ \frac{1}{21} & -\frac{4}{21} \end{bmatrix} = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{1}{21} & -\frac{4}{21} \end{bmatrix}$$

E.g.2:

$$B = \begin{bmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{bmatrix}$$

$$|B| = \sin^2\theta - (-\cos^2\theta) = \sin^2\theta + \cos^2\theta = 1$$

$$B^{-1} = \begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$$

(3 x 3) *and Bigger Dimension Case*

1. Let *A* be an (*n* x *n*) matrix (square matrix).

- The minor of the element a_{ij} of *A* is $\left|A_m\right|_{ij} =$ the determinant of matrix [order (*n* - 1) x (*n* - 1)] with row *i* and they are column *j* deleted.

- The cofactor of the element of *A* is

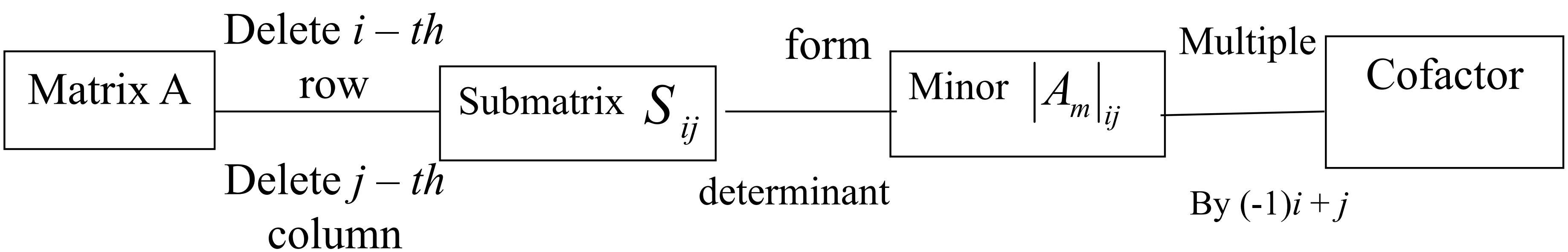
$$\left|A_c\right|_{ij} = (-1)^{i+j}(A_m)_{ij}$$

- The cofactor and minor are coincident if *i* + *j* is even and they are opposite sign if *i* + *j* is add.

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

2. The cofactor matrix of *A* is a matrix whose elements are all cofactor.

3.



(3 x 3) *and Bigger Dimension Case*(*Cont'*)

E.g.:

$$A = \begin{bmatrix} 12 & 6 & -9 \\ 3 & 8 & 15 \\ 4 & 11 & 5 \end{bmatrix}$$

$$|A_m|_{32} = \begin{vmatrix} 12 & -9 \\ 3 & 15 \end{vmatrix} = 207$$

$$|A_m|_{33} = \begin{vmatrix} 12 & 6 \\ 3 & 8 \end{vmatrix} = 78$$

$$|A_c|_{31} = (-1)^{3+1}\begin{vmatrix} 6 & -9 \\ 8 & 15 \end{vmatrix} = (-1)^4\left[(6\cdot 15) - ((-9)\cdot 8)\right] = 90 + 72 = 162$$

$$|A_c|_{32} = (-1)^{3+2}\begin{vmatrix} 12 & -9 \\ 3 & 15 \end{vmatrix} = -\left[(12\cdot 15) - ((-9)\cdot 3)\right] = -(180 + 27) = -207$$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

4. Determinant of matrix *A*

If all element in a given row or column of a square matrix are multiple by their cofactor, the sum of these product is the determinant of the matrix.

$$|A| = a_{i1}|A_c|_{i1} + a_{i2}|A_c|_{i2} + \cdots + a_{in}|A_c|_{in}$$ (select row)

$$|A| = a_{1j}|A_c|_{1j} + a_{2j}|A_c|_{2j} + \cdots + a_{nj}|A_c|_{nj}$$ (select column)

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

- E.g:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{bmatrix}$$

$$|A| = 1 \cdot \begin{vmatrix} 5 & 6 \\ -8 & 9 \end{vmatrix} - 2 \cdot \begin{vmatrix} -4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \cdot \begin{vmatrix} -4 & 5 \\ 7 & -8 \end{vmatrix} = (45-48) - 2(-36-42) + 3(32-35) = 240$$

Expand from first row

$$|A| = -2 \cdot \begin{vmatrix} -4 & 6 \\ 7 & 9 \end{vmatrix} + 5 \cdot \begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} + 8 \cdot \begin{vmatrix} 1 & 3 \\ -4 & 6 \end{vmatrix} = 240$$

Expand from second column

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{bmatrix}$$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

5. In calculating , we select row or column in A with most zero element to simplify the calculation.

E.g.:

$$\begin{bmatrix} 1 & 2 & 3 \\ \boxed{4 \quad 0 \quad 0} \\ 7 & -8 & 9 \end{bmatrix} \quad |A| = -4\begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} + 0 + 0 = -4(18+24) = -4(42) = -168$$

6. The sign of the cofactors follow the pattern below:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \cdots & \cdots & \cdots & \cdots \end{bmatrix}$$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

7. For determinant of matrix whose order is more than (3 x 3):

E.g.:

$$\begin{bmatrix} 3 & 5 & -2 & 6 \\ 1 & 2 & -1 & 1 \\ 2 & 4 & 1 & 5 \\ 3 & 7 & 5 & 3 \end{bmatrix}$$

$$|A| = 3\begin{vmatrix} 2 & -1 & 1 \\ 4 & 1 & 5 \\ 7 & 5 & 3 \end{vmatrix} - 5\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & 5 \\ 3 & 5 & 3 \end{vmatrix} + (-2)\begin{vmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 3 & 7 & 3 \end{vmatrix} - 6\begin{vmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \\ 3 & 7 & 5 \end{vmatrix} = \cdots = -18$$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

8. Inverse of matrix (square matrix)

- The adjoint of *A* is the transpose of the cofactor matrix of *A*.

$$AdjA = \begin{bmatrix} |A_c|_{11} & |A_c|_{12} & \cdots & |A_c|_{1n} \\ |A_c|_{21} & |A_c|_{22} & \cdots & |A_c|_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ |A_c|_{n1} & |A_c|_{n2} & \cdots & |A_c|_{nn} \end{bmatrix}^T$$

- The inverse of *A* is

$$A^{-1} = \frac{1}{|A|} adjA$$

- If

$$|A| \begin{cases} = 0 \text{ , } A \text{ is singular.} \\ \neq 0 \text{ , } A \text{ is non-singular (for invertible).} \end{cases}$$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

- E.g.:

$$A = \begin{bmatrix} 2 & 3 & 0 \\ -5 & 0 & 4 \\ 0 & 2 & 1 \end{bmatrix} \longleftarrow \qquad |A| = 2(-8) - 3(-5) + 0 = -1$$

$m_{ij} = \text{minor}$		$c_{ij} = \text{cofactor}$
$m_{11} = \begin{vmatrix} 0 & 4 \\ 2 & 1 \end{vmatrix} = -8$	+	$c_{11} = m_{11} = -8$
$m_{12} = \begin{vmatrix} -5 & 4 \\ 0 & 1 \end{vmatrix} = -5$	-	$c_{12} = -m_{12} = 5$
$m_{13} = \begin{vmatrix} -5 & 0 \\ 0 & 2 \end{vmatrix} = -10$	+	$c_{13} = m_{13} = -10$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

$m_{21} = \begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix} = 3$	-	$c_{21} = m_{21} = -3$
$m_{22} = \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = 2$	+	$c_{22} = m_{22} = 2$
$m_{23} = \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix} = 4$	-	$c_{23} = m_{23} = -4$
$m_{31} = \begin{vmatrix} 3 & 0 \\ 0 & 4 \end{vmatrix} = 12$	+	$c_{31} = m_{31} = 12$
$m_{32} = \begin{vmatrix} 2 & 0 \\ -5 & 4 \end{vmatrix} = 8$	-	$c_{32} = m_{32} = -8$
$m_{33} = \begin{vmatrix} 2 & 3 \\ -5 & 0 \end{vmatrix} = 15$	+	$c_{33} = m_{33} = 15$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

- The cofactor matrix of *A* is $\begin{bmatrix} -8 & 5 & -10 \\ -3 & 2 & -4 \\ 12 & -8 & 15 \end{bmatrix}$.

- The adjoint of *A* is $\begin{bmatrix} -8 & -3 & 12 \\ 5 & 2 & -8 \\ -10 & -4 & 15 \end{bmatrix}$.

- The inverse of *A*,

$$A^{-1} = \frac{1}{|A|} adjA = -\begin{bmatrix} -8 & -3 & 12 \\ 5 & 2 & -8 \\ -10 & -4 & 15 \end{bmatrix} = \begin{bmatrix} 8 & 3 & -12 \\ -5 & -2 & 8 \\ 10 & 4 & -15 \end{bmatrix}.$$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

- E.g.:

$$A = \begin{bmatrix} -3 & 1 & 2 \\ -5 & -1 & 1 \\ 2 & 1 & -3 \end{bmatrix}$$

$$|A| = \ldots = -25 \neq 0 \Rightarrow A^{-1} \text{ exist}$$

(3 x 3) *and Bigger Dimension Case*(*Cont'*)

- E.g.:

$$B = \begin{bmatrix} -3 & 1 & 2 \\ 1 & 2 & 1 \\ 4 & 1 & -1 \end{bmatrix}$$

$$|B| = \ldots = 0$$

- $\therefore B$ is singular $\Rightarrow B^{-1}$ does not exist.

Some Properties of Determinants, Adj, Inverse

(a) Transpose $\left|A^T\right| = |A|$

- This just states that expanding by the first row or the first column gives the same results.

(b) Product $|AB| = |A||B|$

- This result is difficult to prove generally, but it can be verified rather tediously for 2x2 or 3x3 cases.

For the 2x2 case:

$$|A||B| = (a_{11}a_{22} - a_{12}a_{21})(b_{11}b_{22} - b_{12}b_{21}) = a_{11}a_{22}b_{11}b_{22} - a_{11}a_{22}b_{12}b_{21} - a_{12}a_{21}b_{11}b_{22} + a_{12}a_{21}b_{12}b_{21}$$

Some Properties of Determinants, Adj, Inverse(*Cont'*)

(b)

$$|AB| = \begin{vmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{vmatrix}$$

$$= (a_{11}b_{11} + a_{12}b_{21})(a_{21}b_{12} + a_{22}b_{22}) - (a_{11}b_{12} + a_{12}b_{22})(a_{21}b_{11} + a_{22}b_{21})$$

$$= a_{11}a_{22}b_{11}b_{22} - a_{11}a_{22}b_{12}b_{21} - a_{12}a_{21}b_{11}b_{22} + a_{12}a_{21}b_{12}b_{21}$$

(*c*) $|kA| = k^n |A|$ if *A* is (*n* x *n*) matrix.

(*d*) If a row (column) of *A* consists entirely of zero, then $\det(A) = 0.$

Some Properties of Determinants, Adj, Inverse(Cont')

(e) If two rows (or columns) of matrix *A* are equal (are the same), then $|A| = 0.$ (*A* is singular)

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{21} & a_{22} & a_{23} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{22} & a_{23} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{21} & a_{23} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{21} & a_{22} \end{vmatrix} = 0$$

(*f*) If matrix *B* formed from *A* by multiplying row *i* (or column *j*) by a scalar λ , then $|B| = \lambda|A|.$

$$|B| = \begin{vmatrix} \lambda a_{11} & \lambda a_{12} & \lambda a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \lambda|A|$$

$$A \xrightarrow{R_i : R_i -> \lambda R_i} B \Rightarrow |B| = \lambda|A|$$

Some Properties of Determinants, Adj, Inverse(Cont')

(g) If matrix B formed from matrix A by interchanging two rows (or columns), then $|B| = -|A|$.

-Consider $|A|$ and $|B|$ in which row 1 and 2 are interchanged:

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \qquad |B| = \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

-Expanding $|A|$ by the first row,

$$|A| = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Some Properties of Determinants, Adj, Inverse(*Cont'*)

and $|B|$ by the second row,

$$|B| = -a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} - a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Thus $|A| = -|B|$ so that interchanging two rows change the sign of the determinant. Entirely similar results apply changing two columns.

$$A \xrightarrow{R_i <-> R_j} B \Rightarrow |B| = -|A|$$

Some Properties of Determinants, Adj, Inverse(*Cont'*)

(*h*) If *B* formed from *A* by adding(λ times row *i*) to row *k*, then $|B| = |A|$.

$$A \xrightarrow{R_k : R_k + \alpha R_i} B \Rightarrow |B| = |A|$$

(*i*) $|A^{-1}| = \dfrac{1}{|A|}$

(*j*) $AA^{-1} = A^{-1}A = I$

(k) $AB = I \Rightarrow B = A^{-1}$

(l) $adj(AB) = adj(B)adj(A)$

(m) $|I| = 1$ Determinant for identity matrix

Some Properties of Determinants, Adj, Inverse(*Cont'*)

- E.g.:

Evaluate the 3 x 3 determinants:

(*a*) $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 0 \end{vmatrix}$ by expanding the first row.

(*b*) $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \end{vmatrix}$ by expanding the first column.

(*c*) $\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{vmatrix}$ by expanding the third row.

(*d*) $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 4 \\ 3 & 3 & 0 \end{vmatrix}$

Some Properties of Determinants, Adj, Inverse(*Cont'*)

- Solution:

(a) $$\begin{vmatrix}1 & 0 & 1\\0 & 1 & 2\\1 & 1 & 0\end{vmatrix} = 1\begin{vmatrix}1 & 2\\1 & 0\end{vmatrix} - 0\begin{vmatrix}0 & 2\\1 & 0\end{vmatrix} + 1\begin{vmatrix}0 & 1\\1 & 1\end{vmatrix} = -2-0-1=-3$$

(b) $$\begin{vmatrix}1 & 0 & 1\\1 & 1 & 0\\0 & 1 & 2\end{vmatrix} = 1\begin{vmatrix}1 & 0\\1 & 2\end{vmatrix} - 1\begin{vmatrix}0 & 1\\1 & 2\end{vmatrix} + 0\begin{vmatrix}0 & 1\\1 & 0\end{vmatrix} = 2+1+0=3$$

Note that (a) and (b) are the same determinant, but with two rows interchanged. The result confirms property (c) just stated above.

Some Properties of Determinants, Adj, Inverse(Cont')

(c) $$\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{vmatrix} = 1\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} - 0\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + 2\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1 - 0 + 2 = 3$$

Note that the matrix associated with the determinant in (c) is just the transpose of matrix associated with the determinant in (b).

(d) $$\begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 4 \\ 3 & 3 & 0 \end{vmatrix} = 2\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 3 & 3 & 0 \end{vmatrix} = 2(3)\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 0 \end{vmatrix} = 2(3)(-3) = -18$$

Note that we have used the multiple of a row rule on two occasions, the final determinant is the same as (a).

Some Properties of Determinants, Adj, Inverse(Cont')

- E.g: Making use of property (*f*), (*g*), (*h*) of determinant:

$$\begin{vmatrix} -6 & 0 & 1 & 3 & 2 \\ -1 & 5 & 0 & 1 & 7 \\ 8 & 3 & 2_{33} & 1 & 7 \\ 0 & 1 & 5_{43} & -3 & 2 \\ 1 & 15 & -3_{53} & 9 & 4 \end{vmatrix} = \begin{vmatrix} -6 & 0 & 1 & 3 & 2 \\ -1 & 5 & 0 & 1 & 7 \\ 20 & 3 & 0 & -5 & 3 \\ 30 & 1 & 0 & -18 & -8 \\ -17 & 15 & 0 & 18 & 10 \end{vmatrix}$$

$$\boxed{R_3 : R_3 + (-2)R_1}$$

$$\boxed{R_4 : R_4 + (-5)R_1}$$

$$\boxed{R_5 : R_5 + 3R_1}$$

Some Properties of Determinants, Adj, Inverse(Cont’)

- Expand column 3 (Most zero column)

$$
\begin{vmatrix}
\cancel{-6} & \cancel{0} & \cancel{1} & \cancel{3} & 2 \\
-1 & 5 & \cancel{0} & 1 & 7 \\
20 & 3 & \cancel{0} & -5 & 3 \\
30 & 1 & \cancel{0} & -18 & -8 \\
-17 & 15 & \cancel{0} & 18 & 10
\end{vmatrix}
=
\begin{vmatrix}
-1 & 5_{12} & 1_{13} & 7_{14} \\
20 & 3 & -5 & 3 \\
30 & 1 & -18 & -8 \\
-17 & 15 & 18 & 10
\end{vmatrix}
=
\begin{vmatrix}
-1 & 0 & 0 & 0 \\
20 & 103 & 15 & 143 \\
30 & 151 & 12 & 204 \\
-17 & -70 & 1 & -109
\end{vmatrix}
$$

$$\boxed{C_2 : C_2 + 5C_1}$$

$$\boxed{C_3 : C_3 + C_1}$$

$$\boxed{C_4 : C_4 + 7C_1}$$

Some Properties of Determinants, Adj, Inverse(Cont')

- Expand row 1(Most zero row)

$$\begin{vmatrix} -1 & 0 & 0 & 0 \\ 20 & 103 & 15 & 143 \\ 30 & 151 & 12 & 204 \\ -17 & -70 & 1 & -109 \end{vmatrix} = -\begin{vmatrix} 103 & 15 & 143 \\ 151 & 12 & 202 \\ -70 & 1 & -109 \end{vmatrix} = -\begin{vmatrix} 1153 & 0 & 1778 \\ 991 & 0 & 1510 \\ -70 & 1 & -109 \end{vmatrix}$$

$$\boxed{R_1 : R_1 + (-15)R_3}$$

$$\boxed{R_2 : R_2 + (-12)R_3}$$

Some Properties of Determinants, Adj, Inverse(Cont')

- Expand column 2(Most zero column)

$$-\begin{vmatrix} 1153 & 0 & 1778 \\ 991 & 0 & 1510 \\ -70 & 1 & -109 \end{vmatrix} = -(-1)\begin{vmatrix} 1153 & 1778 \\ 991 & 1510 \end{vmatrix} = -20968$$

Trace of a Matrix (Square Matrix)

1. Trace of matrix *M*,

   $Tr(M) = \sum_{i=1}^{n} m_{ii} =$ the summation of all the diagonal elements.

E.g:

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 6 \\ 4 & 3 & 8 \end{bmatrix} \qquad Tr(A) = 1 + 0 + 8 = 9$$

2. If *A* and *B* are square matrices with real (or complex) entries, of same order and *c* is scalar, then

$$Tr(A+B) = Tr(A) + Tr(B)$$
$$Tr(cA) = cTr(A)$$

Trace of a Matrix (Square Matrix)(Cont')

3. If *A* and *B* are matrices such that *AB* is square matrix, then $Tr(AB) = Tr(BA)$.

4. If *A*, *B*, *C* are matrices such that *ABC* is square matrix, then $Tr(ABC) = Tr(CAB) = Tr(BCA)$.

Trace of a Matrix (Square Matrix)(Cont')

- We do not have the matrix-matrix division, but we can have because $A^{-1}A^{n}$

$$
\begin{aligned}
&A^{-1}A^{n} \\
&= A^{-1}\left(AA^{n-1}\right) \\
&= \left(A^{-1}A\right)A^{n-1} \\
&= IA^{n-1} \\
&= A^{n-1}
\end{aligned}
$$

- i.e. Multiple by the inverse of *A* will reduce its power to *n* -1, nevertheless we do not divide A^{n} by *A*.

Negative Power

- : $A^{-n} = \left(A^{-1}\right)^n = \left(A^n\right)^{-1}$
- $\Rightarrow A^{-n}$ is the inverse of or is the inverse of *A* to the power of *n*.

*Elementary **ROW** Operation*

1. Interchange two rows: $R_i \leftrightarrow R_j$

2. Multiply a row by a non-zero scalar *k*: $R_i \rightarrow kR_i$

3. Add a multiple of one row to another row:

$$R_i \rightarrow R_i + hR_j$$

- We can combine 2. and 3. : $R_i \rightarrow kR_i + hR_j$

E.g:

$$\begin{bmatrix} 2 & 1 & 5 \\ 3 & 4 & 0 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_2 : R_2 \leftrightarrow R_3} \begin{bmatrix} 2 & 1 & 5 \\ 7 & 8 & 9 \\ 3 & 4 & 0 \end{bmatrix} \xrightarrow{R_1 : 3R_1} \begin{bmatrix} 6 & 3 & 15 \\ 7 & 8 & 9 \\ 3 & 4 & 0 \end{bmatrix} \xrightarrow{R_1 : R_1 - 2R_3} \begin{bmatrix} 0 & -5 & 15 \\ 7 & 8 & 9 \\ 3 & 4 & 0 \end{bmatrix}$$

Elementary ***ROW*** *Operation (Cont')*

- E.g.:

Using elementary row operations, transform *A* into *B*.

$$A=\begin{bmatrix} 2 & 1 & -1 & 3 \\ 1 & 2 & 1 & 3 \\ 3 & 1 & 0 & 2 \end{bmatrix} \qquad B=\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & -5 & -3 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & -1 & 3 \\ 1 & 2 & 1 & 3 \\ 3 & 1 & 0 & 2 \end{bmatrix} \xrightarrow{\boxed{R_1 : R_1 \leftrightarrow R_2}} \begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & -1 & 3 \\ 3 & 1 & 0 & 2 \end{bmatrix} \xrightarrow{\boxed{R_2 : R_2 - 2R_1}}$$

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & \boxed{-3 \quad -3 \quad -3} \\ 3 & 1 & 0 & 2 \end{bmatrix} \xrightarrow{\boxed{R_3 : R_3 - 3R_1}} \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & -3 & -3 & -3 \\ 0 & \boxed{-5 \quad -3 \quad -7} \end{bmatrix}$$

$$\xrightarrow{\boxed{R_2 : -\frac{1}{3} R_2}} \begin{bmatrix} 1 & 2 & 1 & 3 \\ \boxed{0 \quad 1 \quad 1 \quad 1} \\ 0 & -5 & -3 & -7 \end{bmatrix}$$

Row Echelon Matrix

- *Properties of matrices in echelon form:*

1. If a row does not consist entirely of zeros, then the first non-zero number in the row is 1. We call this as “leading 1”.

2. If there are any rows that consist entirely of zeros, then they are grouped together at the bottom of the matrix.

3. In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row.

Row Echelon Matrix

Properties of matrices in echelon form: (*Cont'*)

4. Each column that contains a leading 1 has zeros everywhere else in that column(other elements in a column 1 must be zero).

- **A matrix has the all 4 properties is said to be in "reduced-row echelon form".
- **A matrix that has the first 3 properties is said to be in "row echelon form".

Row Echelon Matrix (Cont')

- Draw a ladder, top to bottom, left to right, only can go down **ONE** step **vertically**, but can go many steps right horizontally.

- The first element on a ladder must be unity, all elements vertically below must be zero.

Row Echelon Matrix(Cont')

- E.g:

$$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 4 & -3 & 4 \\ 0 & 1 & 6 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

R, RR   R   R, RR   R, RR

- *Note*: *A matrix in row-echelon form had zeros below each leading* 1*, whereas a matrix in reduces row-echelon form has zeros below and above each leading* 1*.*

Rank of a Matrix

1. We can transform a matrix to row-echelon form by (a series of) elementary row operation.

2. Rank, r = the number of non-zero rows in the matrix after it has been transformed to row-echelon form via elementary row operation.

Rank of a Matrix(Cont')

- E.g:

Change to row-echelon form and find the rank.

$$\begin{bmatrix} 1 & 2 & -1 & 6 \\ 3 & 8 & 9 & 10 \\ 2 & -1 & 2 & -2 \end{bmatrix} \xrightarrow{R_2:R_2-3R_1} \begin{bmatrix} 1 & 2 & -1 & 6 \\ 0 & 2 & 12 & -8 \\ 2 & -1 & 2 & -2 \end{bmatrix} \xrightarrow{R_3:R_3-2R_1} \begin{bmatrix} 1 & 2 & -0 & 6 \\ 0 & 2 & 12 & -8 \\ 0 & -5 & 4 & -14 \end{bmatrix} \xrightarrow{R_2:\frac{1}{2}R_2}$$

$$\begin{bmatrix} 1 & 2 & -1 & 6 \\ 0 & 1 & 6 & -4 \\ 0 & 0 & 1 & -1 \end{bmatrix} \xleftarrow{R_3:\frac{1}{34}R_3} \begin{bmatrix} 1 & 2 & -1 & 6 \\ 0 & 1 & 6 & -4 \\ 0 & 0 & 34 & -34 \end{bmatrix} \xleftarrow{R_3:R_3+5R_2} \begin{bmatrix} 1 & 2 & -1 & 6 \\ 0 & 1 & 6 & -4 \\ 0 & -5 & 4 & -14 \end{bmatrix}$$

There are 3 non-zero rows (all rows are non-zero), so the rank of the matrix is 3.

Application of Matrix to Solve System of Simulation Linear Equation

- *Linear Equation*

1. A linear equation with n unknown has the following general form

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b$$

where

- $a_1, a_2, \dots, a_n$ are coefficients
- $x_1, x_2, \dots, x_n$ are unknowns
- b is a known constant.

- *Linear Equation(Cont')*

2. The equation is linear because all variable (the unknowns) $x_1, x_2, \ldots, x_n$ has degree one.

3. A solution of the equation is any sequence $s_1, s_2, \ldots, s_n$ of numbers such that the substitution $x_1 = s_1, x_2 = s_2, \ldots, x_n = s_n$ satisfied the equation.

- *Linear Equation(Cont')*
- E.g: $x_1 + 2x_2 + x_3 = 1$

One possible solution (or solution set) is $x_1 = 2, x_2 = 1, x_3 = -3$.

- E.g:

$$\left.\begin{aligned} x_1 + 2x_1x_2 + 3x_3 &= 5 \\ x_1^2 + 3\sqrt{x_2} + \sin(x_3) &= 7 \\ \frac{1}{x_1} + 7x_2^{-2} &= 0 \end{aligned}\right\} \text{Nonlinear equation}$$

- E.g:

$$\begin{aligned} x_1 - x_2 + x_3 &= 3 \\ x_1 + 5x_2 - 5x_3 &= 2 \\ 2x_1 + x_2 - x_3 &= 1 \end{aligned} \qquad \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 1 & 5 & -5 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right] \quad or \quad \begin{bmatrix} 1 & -1 & 1 \\ 1 & 5 & -5 \\ 2 & 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

System of Linear Equation: Definition and Notation

1. A linear system of *m* equations in *n* unknowns is a system of the form

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nn}x_n &= b_m
\end{aligned}
$$

where a_{ij}'s and b_i's are real numbers.

- The above system is an (*n* x *n*) linear system, i.e, the dimension of the system is (*m* x *n*).

System of Linear Equation: Definition and Notation(Cont')

- $a_{ij}'s$ are coefficients.

- $b_i's$ are known constant (could be zero).

- $x_i's$ are unknown.

- A solution (solution set) to the system $s_1, s_2, \ldots, s_n$ of numbers that simultaneously satisfies each equation in the system.

System of Linear Equation: Definition and Notation(Cont')

- E.g. 1:

(2 x 2) linear system

$$x_1 + 2x_2 = 5$$
$$2x_1 + 3x_2 = 8$$

$x_1 = 1,\ x_2 = 2$ is a solution to the system.

E.g.2:

$$x_1 - x_2 + x_3 = 2$$
$$2x_1 + x_2 - x_3 = 4$$

- This is a (2 x 3) linear system.
- The order triple (2, 0, 0) is a solution.

System of Linear Equation: Definition and Notation(Cont')

- If the constant $b_1, b_2, \ldots, b_n$ are $\boxed{\text{all zero}}$, the system is $\boxed{\text{homogenous}}$, otherwise, it is non-homogeneous.

- A system containing as many equations as unknown is called square, i.e, $m = n$.

- The system may have

i. one (unique) solution => the system is determinate (and also consistent)
ii. No solution => the system is inconsistent.
iii. infinite many solution => the system is consistent.

General Interpretations of Solutions Sets: (2 x 2) *system*

1. $$a_{11}x_1 + a_{12}x_2 = b_1$$
   $$a_{21}x_1 + a_{22}x_2 = b_2$$

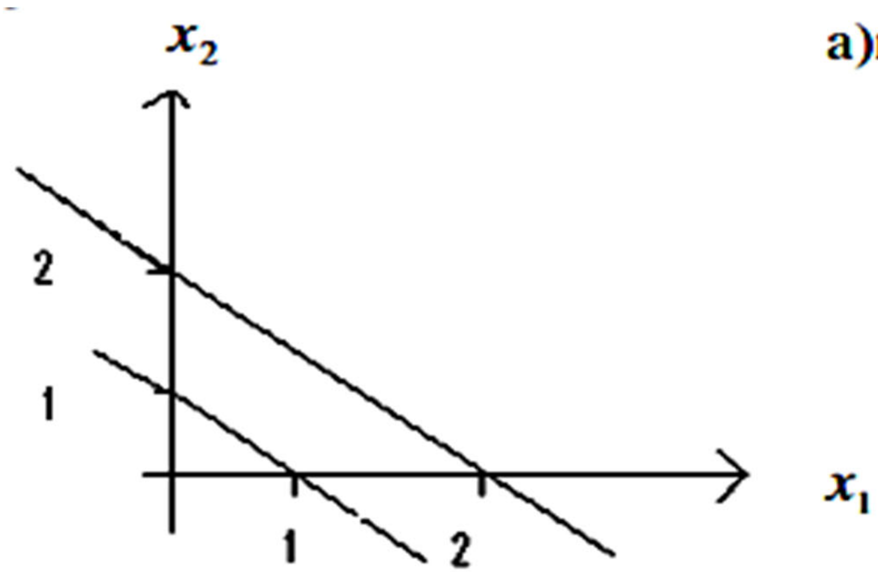
   $a_{ij}'s$ where all are non-zero.

- The equation can be represented a lines in the planes (2-D plane)
- The solution of the system (x_1, x_2) corresponds to the point where the lines interests.

General Interpretations of Solutions Sets: (2 x 2) *system*

2. When the system has

a) no solution => The two lines are parallel (never meet)

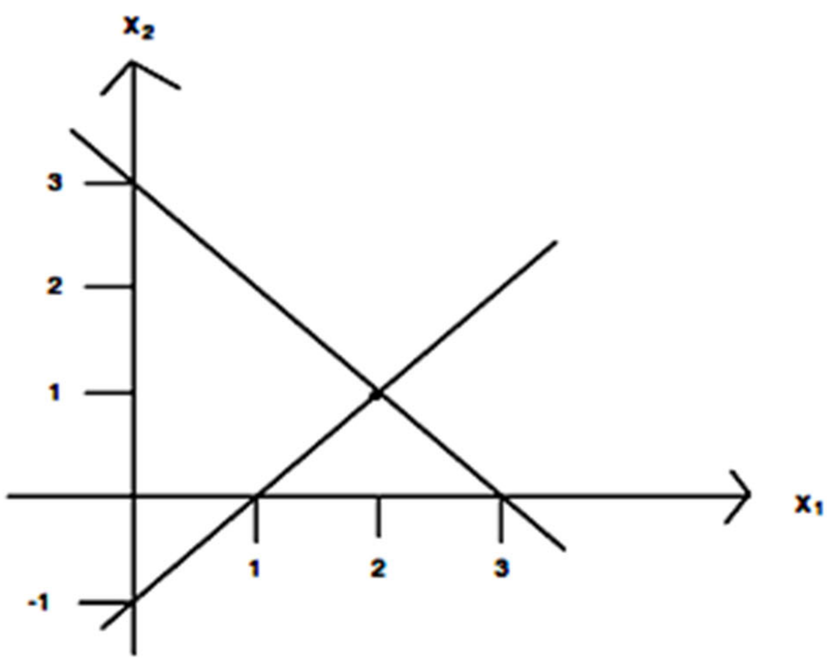


a)no solution

$$x_1 + x_2 = 2$$
$$x_1 + x_2 = 1$$

General Interpretations of Solutions Sets: (2 x 2) *system*

b) one solution (unique solution) => The two lines intersect at a single point



b)one solution

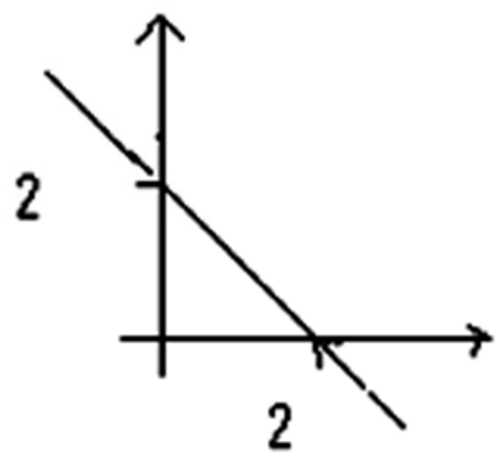
$$x_1 + x_2 = 3$$

$$x_1 - x_2 = 1$$

(2,1) is a solution.

General Interpretations of Solutions Sets: (2 x 2) *system*

c) infinite many solution => The two lines are coincident (at the same line), their intersection is a line (the same line).



c)infinite many solution

$$x_1 + x_2 = 2$$
$$2x_1 + 2x_2 = 4$$

** If the constant b_1, b_2 all zero, all the line passes through the origin.

General Interpretations of Solutions Sets: (2 x 2) *system*

3. For (3 x 3) system, each linear equation of the system corresponds to a plane.

4. For dimension higher then (3 x 3), we do not consider their geometric interpretation.

Equivalent System

1. Two systems of equations involving the same unknowns (variables) are said to be equivalent if they have the same solution set.

- E.g:

$$\begin{aligned} 3x+2y-z&=-2 \\ y&=3 \\ 2z&=4 \end{aligned} \quad \text{and} \quad \begin{aligned} 3x+2y-z&=-2 \\ -3x-y+z&=5 \\ 3x+2y+z&=2 \end{aligned} \quad \text{are equivalent,}$$

they share a common solution set (-2, 3, 2).

Matrix Representation of a Linear System; Augmented Matrix

1. Consider the (m x n) system

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\
\vdots \qquad\qquad\qquad\qquad & \quad \vdots \\
a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m
\end{aligned}$$

2. This system can be rewritten in the matrix form

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

or, with appropriate designations for the matrices: $Ax = b$

Matrix Representation of a Linear System; Augmented Matrix(Cont')

2. A = matrix of coefficients

   x = the column vector of unknown

   b = the column vector of constants.

3. We can juxtapose b with A to form an augmented matrix (or partition matrix)

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] = \left[A \middle| b\right]$$

Matrix Representation of a Linear System; Augmented Matrix(Cont')

4. It is possible to perform elementary row operations upon the augmented matrix without violating the given equations.

5. If

$$b \begin{cases} = 0 \Rightarrow \text{homogeneous system} \\ \neq 0 \Rightarrow \text{non-homogeneous system} \end{cases}$$

6. We can solve a linear system by matrix method
   a. Gaussian elimination method
   b. Inverse of matrix *A*
   c. Cramer's Rule

Matrix Representation of a Linear System; Augmented Matrix(Cont')

- E.g:

$$\begin{aligned} x_1 - x_2 + x_3 &= 3 \\ x_1 + 5x_2 - 5x_3 &= 2 \\ 2x_1 + x_2 - x_3 &= 1 \end{aligned} \rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 1 & 5 & -5 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \xrightarrow[\text{Augmentd matrix}]{} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 1 & 5 & -5 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right]$$

Gaussian Elimination Method & Gauss Jordan Elimination

1. The method is carried out by applying elementary row operation to the augmented matrix to transform it to row-echelon form.

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

Gaussian Elimination Method & Gauss Jordan Elimination(Cont')

2. The form produce of the method are:

i. Divide the first row by and then subtract suitable multiples of the first row from the subsequent row, to obtain a matrix of the form.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

(If $a_{11} = 0$, interchange two rows first).

Gaussian Elimination Method & Gauss Jordan Elimination(Cont')

ii. The first row now remains untouched and the process is repeated with the remaining rows. That is, divided the second row by to produce a 1, and subtract suitable multiple of the new second from the subsequent rows to produce zeros below that 1.

iii. The process continues in the same way until the row-echelon form is reach.

iv. Determine the rank of the augmented matrix.

v. Determine the number of arbitrary parameters (independent variables) need to de set.

vi. Solve the system of equation by back substitution method.

Gaussian Elimination Method & Gauss Jordan Elimination(Cont')

3. We can conclude whether the system has one solution, infinite many solutions or no solution by

- looking at the form of the augment matrix in echelon form
- checking the value of $|A|$, the determinant of *A*

**checking/concluding the type of solutions by looking at

$\rightarrow |A|$

$\rightarrow$ the form of augmented matrix

Checking/Concluding the Type of Solution

1.

- $Ax = b$
  - $b = 0$ (homogeneous)
    - $|A| \neq 0$, trivial solution, $x_i = 0, \forall i$
    - $|A| = 0$ infinite many solution
  - $b \neq 0$ (Non-homogeneous)
    - $|A| \neq 0$, unique solution (one solution)
    - $|A| = 0$
      - No solution, system inconsistent
      - Infinite many solutions

Checking/Concluding the Type of Solution

2.

Homogeneous

$$\left[\begin{array}{ccc|c} x & x & x & 0 \\ 0 & x & x & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} x & x & x & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$n > r$
Infinite many solutions

$$\left[\begin{array}{ccc|c} x & x & x & 0 \\ 0 & x & x & 0 \\ 0 & 0 & x & 0 \end{array}\right]$$

trivial solution, $x_i = 0, \forall i$

Non-Homogeneous

$$\left[\begin{array}{ccc|c} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} x & x & x & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$n > r$
Infinite many solutions

$$\left[\begin{array}{ccc|c} x & x & x & 0 \\ 0 & x & x & x \\ 0 & 0 & 0 & x \end{array}\right]$$

No solution, system inconsistent

$$\left[\begin{array}{ccc|c} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right]$$

unique solution (one solution)

Example 1

Solve the following system of linear equation by Gaussian Elimination Method.

$$\begin{aligned} x + y &= 2 \\ x - y &= 2 \end{aligned}$$

- First reduce the augmented matrix to row-echelon form.

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & -1 & 2 \end{array}\right] \xrightarrow{\boxed{R_2 : R_2 - R_1}} \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -2 & 0 \end{array}\right] \xrightarrow{\boxed{R_2 : -\frac{1}{2} R_2}} \underset{\text{REM}}{\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 0 \end{array}\right]}$$

$$\begin{aligned} x + y &= 2 \\ y &= 0 \end{aligned}$$

- By back substitution, we obtain x = 2. This gives

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

Example 2

$$x - y - 5z = -1$$
$$-2x + 2y + 11z = 1$$
$$3x - y + z = 3$$

The augmented matrix for the system is

$$\left[\begin{array}{ccc|c} 1 & -1 & -5 & -1 \\ -2 & 2 & 11 & 1 \\ 3 & -1 & 1 & 3 \end{array}\right] \xrightarrow{R_2:R_2+2R_1} \left[\begin{array}{ccc|c} 1 & -1 & -5 & -1 \\ 0 & 0 & 1 & -1 \\ 3 & -1 & 1 & 3 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & -1 & -5 & -1 \\ 3 & -1 & 1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow{R_2:R_2-3R_1} \left[\begin{array}{ccc|c} 1 & -1 & -5 & -1 \\ 0 & 2 & 16 & 6 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

$$\xrightarrow{R_2:\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & -1 & -5 & -1 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right]$$ Row-echelon form

To finish by ***<u>Gaussian</u>***, using back substitution,

$$z = -1$$
$$y + 8z = 3$$
$$y = 3 - 8z = 3 - 8(-1) = 3 + 8 = 11$$
$$x - y - 5z = -1$$
$$x = -1 + 5z + y = -1 + 5(-1) + 11 = 5$$

$r = 3$, $n = 3$,

degree of freedom, $d.o.f = n - r = 0$

Determinant $\neq 0$, $b \neq 0$

unique solution.

Example 2(cont')

- To finish by ***Gauss-Jordan***,

$$\left[\begin{array}{ccc|c} 1 & -1 & -5 & -1 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow{R_1:R_1+R_2} \left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow{R_2:R_2-8R_3} \left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & 0 & 11 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow{R_1:R_1-3R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 11 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

Row-reduced echelon form

Hence, we get $x = 5, y = 11, z = -1$.

Example 3

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 2 \\ 8 & 3 & 5 & 4 \\ 3 & 1 & 2 & -2 \end{array}\right] \xrightarrow{R_1:\frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 1 \\ 8 & 3 & 5 & 4 \\ 3 & 1 & 2 & -2 \end{array}\right] \xrightarrow{R_2:R_2-8R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & -1 & 1 & -4 \\ 3 & 1 & 2 & -2 \end{array}\right] \xrightarrow{R_2:-R_2} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 1 & -1 & 4 \\ 3 & 1 & 2 & -2 \end{array}\right]$$

$$\xrightarrow{R_3:R_3-3R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 1 & -1 & 4 \\ 0 & -\frac{1}{2} & \frac{1}{2} & -5 \end{array}\right] \xrightarrow{R_3:R_3+\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

- Check: determinant = 0
- rank of the coefficient matrix = 2 and the rank of the augmented matrix = 3
- No solution, the linear system $Ax = b$ is NOT consistent.

Example 4

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & 3 & 1 & 6 \\ 5 & 7 & 1 & 13 \end{array}\right] \xrightarrow{R_2:R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & 3 & 4 \\ 5 & 7 & 1 & 13 \end{array}\right] \xrightarrow{R_3:R_3-5R_1} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 2 & 6 & 8 \end{array}\right] \xrightarrow{R_3:R_3-2R_2} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

- -degree of freedom, $d.o.f = n - r = 3 - 2 = 1$ Infinite many solution
- -one arbitrary parameter.
- -determinant = 0

-circle is not free.
-column(s) which has NO leading 1 gives free variables.

- Let $z = t$

$$y + 3z = 4$$

$$y = 4 - 3z = 4 - 3t$$

$$x + y - z = 1$$

$$x = 1 - y + z = 1 - (4 - 3t) + t = 4t - 3$$

Example 5

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array}\right] \to \cdots \to \begin{array}{c} \begin{array}{cccc} w & x & y & z \end{array} \\ \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

- -degree of freedom, $d.o.f = n - r = 4 - 2 = 2$
- Infinite many solution
- two arbitrary parameter.

Set $x = \alpha, z = \beta$

$y - 2z = 1$

$y = 1 + 2z = 1 + 2\beta$

$w - 2x - y + 3z = 1$

$w = 1 + 2x + y - 3z = 1 + 2\alpha + 1 + 2\beta - 3\beta = 2 + 2\alpha - \beta$

Example 6

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 2 & 3 & -2 & 0 \\ 4 & -1 & 5 & 0 \end{array}\right] \xrightarrow{R_2:R_2-2R_1} \left[\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 7 & -8 & 0 \\ 4 & -1 & 5 & 0 \end{array}\right] \xrightarrow{R_3:R_3-4R_1} \left[\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 7 & -8 & 0 \\ 0 & 7 & -7 & 0 \end{array}\right]$$

$$\xrightarrow{R_2:\frac{1}{7}R_2} \left[\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 1 & -\frac{8}{7} & 0 \\ 0 & 7 & -7 & 0 \end{array}\right] \xrightarrow{R_3:R_3-7R_2} \left[\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 1 & -\frac{8}{7} & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\text{Determinant} \neq 0$$

$$z = 0$$

$$y - \frac{8}{7}z = 0$$

$$\Rightarrow \; y = 0$$

$$x - 2y + 3z = 0$$

$$x = 0$$

trivial solution

Example 7

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 2 & 3 & -2 & 0 \\ 4 & -1 & 4 & 0 \end{array}\right] \rightarrow \cdots \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 1 & -\frac{8}{7} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

- degree of freedom, *d.o.f* = *n* – *r* = 3 – 2 = 1 > 0
- one arbitrary parameter.
- Infinite many solution

$$Let$$

$$z = \alpha$$

$$y - \frac{8}{7}z = 0$$

$$y = \frac{8}{7}z = \frac{8}{7}\alpha$$

$$x - 2y + 3z = 0$$

$$x = 2y - 3z = 2\left(\frac{8}{7}\alpha\right) - 3\alpha = \frac{16}{7}\alpha - 3\alpha = -\frac{5}{7}\alpha$$

Example 8

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-R_1} \left[\begin{array}{ccc|c} 1 & -2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

- degree of freedom, *d.o.f* = *n* – *r* = 3 – 1 = 2
- Two arbitrary parameter.
- Infinite many solution

$$Let$$

$$y = \alpha, z = \beta$$

$$x - 2y - z = 0$$

$$x = 2y + z = 2\alpha + \beta$$

Inverse of Matrix Method

1. In general, a system of linear equation can be represent in the following matrix forms

$$AX = b$$

$$A^{-1}AX = A^{-1}b$$

$$IX = A^{-1}b$$

$$X = A^{-1}b$$

Inverse of Matrix Method

1. In general, a system of linear equation can be represent in the following matrix forms

*Multiply both sides by (inverse of A)

Note that $A^{-1}A = AA^{-1} = I$

$$AX = b$$
$$A^{-1}AX = A^{-1}b$$
$$IX = A^{-1}b$$
$$X = A^{-1}b$$

i.e. the solution sets is given by the product of the inverse of *A* and the column vector of constant *b*.

Inverse of Matrix Method (E.g.)

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -3 & 2 \\ 4 & -7 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ -2 \end{bmatrix}$$

$$A \qquad X \qquad b$$

- Check:

$$|A| = \begin{vmatrix} -3 & 2 \\ -7 & 5 \end{vmatrix} - (-2)\begin{vmatrix} 2 & 2 \\ 4 & 5 \end{vmatrix} + 3\begin{vmatrix} 2 & -3 \\ 4 & -7 \end{vmatrix} = -3$$

- Non-homogeneous system
- unique solution

Inverse of Matrix Method (E.g.)

- Cofactor matrix of A:

$$\begin{bmatrix} +\begin{vmatrix} -3 & 2 \\ -7 & 5 \end{vmatrix} & -\begin{vmatrix} 2 & 2 \\ 4 & 5 \end{vmatrix} & +\begin{vmatrix} 2 & -3 \\ 4 & -7 \end{vmatrix} \\ -\begin{vmatrix} -2 & 3 \\ -7 & 5 \end{vmatrix} & +\begin{vmatrix} 1 & 3 \\ 4 & 5 \end{vmatrix} & -\begin{vmatrix} 1 & -2 \\ 4 & -7 \end{vmatrix} \\ +\begin{vmatrix} -2 & 3 \\ -3 & 2 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 2 & 2 \end{vmatrix} & +\begin{vmatrix} 1 & -2 \\ 2 & -3 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} -1 & -2 & -2 \\ -11 & -7 & -1 \\ 5 & 4 & 1 \end{bmatrix}; adjA = \begin{bmatrix} -1 & -11 & 5 \\ -2 & -7 & 4 \\ -2 & -1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} adjA = \frac{1}{3}\begin{bmatrix} 1 & 11 & -5 \\ 2 & 7 & -4 \\ 2 & 1 & -1 \end{bmatrix}$$

$$\therefore X = \frac{1}{3}\begin{bmatrix} 1 & 11 & -5 \\ 2 & 7 & -4 \\ 2 & 1 & -1 \end{bmatrix}\begin{bmatrix} 4 \\ -1 \\ -2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 3 \\ 9 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} \qquad \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$$

Cramer's Rule

- If is a system of linear equations with dimension (n x n) such that $|A| \neq 0,$ then the system has unique solution. This solution is

$$x_1 = \frac{|D_1|}{|A|}, x_2 = \frac{|D_2|}{|A|}, \cdots, x_i = \frac{|D_i|}{|A|}$$

- where D_i is the matrix obtained by replacing its i^{th} column by b.

Cramer's Rule (Cont')

E.g:

$$3x - y = 9$$

$$x + 2y = -4$$

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 9 \\ -4 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$|A| = \begin{vmatrix} 3 & -1 \\ 1 & 2 \end{vmatrix} = 7 \neq 0; \; |D_1| = \begin{vmatrix} \boxed{\begin{matrix} 9 \\ -4 \end{matrix}} & \begin{matrix} -1 \\ 2 \end{matrix} \end{vmatrix} = 14; \; |D_2| = \begin{vmatrix} \begin{matrix} 3 \\ 1 \end{matrix} & \boxed{\begin{matrix} 9 \\ -4 \end{matrix}} \end{vmatrix} = -21$$

$$x = \frac{|D_1|}{|A|} = \frac{14}{7} = 2; \quad y = \frac{|D_2|}{|A|} = \frac{-21}{7} = -3$$

Cramer's Rule (Cont')

E.g:

$$\left[\begin{array}{ccc|c} 2 & 3 & -1 & 4 \\ 3 & 1 & 2 & 13 \\ 1 & 2 & -5 & -11 \end{array}\right] \qquad |A| = \begin{vmatrix} 2 & 3 & -1 \\ 3 & 1 & 2 \\ 1 & 2 & -5 \end{vmatrix} = 28 \neq 0$$

$$|D_1| = \begin{vmatrix} 4 & 3 & -1 \\ 13 & 1 & 2 \\ -11 & 2 & -5 \end{vmatrix} = 56 \qquad |D_2| = \begin{vmatrix} 2 & 4 & -1 \\ 3 & 13 & 2 \\ 1 & -11 & -5 \end{vmatrix} = 28 \qquad |D_3| = \begin{vmatrix} 2 & 3 & 4 \\ 3 & 1 & 13 \\ 1 & 2 & -11 \end{vmatrix} = 84$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$$

Cramer's Rule (Cont')

Limitations

1. Cramer's rule is derived from the inverse of matrix method. They both have some limitation (same limitation).
2. Applicable only when $|A| \neq 0$ (one unique solution system), and for square system, (n x n) system.
3. When the system is not square dimension, has no solution, has infinite solution, the two methods do not help and we have to resort to Gaussian elimination.

   E.g: $\left.\begin{array}{l} x + y = 2z = 6 \\ 4x - y + 5z = 3 \end{array}\right\}$ Cramer's Rule can't help us because *A* is of order (2 x 3), not square, $|A|$ is not defined.
4. When $|A| = 0$, both method will give us divide-by-zero problem.

Thank you ☺